# Data Science
# Capstone Project Report

SpaceX Falcon 9 first-stage landing analysis

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# Executive summary

90 historical Falcon 9 launches produced 60 successful landing outcomes (66.7%).

We collected API and Wikipedia data, cleaned outcomes, explored SQL and visual patterns, mapped launch sites, and trained four classifiers.

All four models scored 83.3% on 18 test launches. The decision tree had the highest training CV accuracy (87.7%).

Use these results for exploratory planning. They do not establish future-launch reliability or launch cost.

https://github.com/rquilo/spacex-capstone-analysis

# Introduction: first-stage landing prediction

Business question: can launch and booster characteristics help predict a successful Falcon 9 first-stage landing?

Reusability motivates the analysis, but landing success alone does not prove recovery, reuse or a specific cost saving.

Target: Class 1 for observed successful landings. Class 0 groups failed, unknown and unattempted outcomes according to the course definition.

https://github.com/rquilo/spacex-capstone-analysis/blob/main/notebooks/03_data_wrangling.ipynb

# Data sources and scope

Each dataset supports a different exercise. Keep its denominator with every conclusion.

| Source | Records | Purpose |
| --- | --- | --- |
| IBM static API response | 107 raw | Collection and filtering |
| Falcon 9 prepared snapshot | 90 | EDA and machine learning |
| Wikipedia historical tables | 121 numbered Falcon 9 | Independent web scraping |
| IBM SQL snapshot | 101 | SQL exploration |
| Geographic / Dash snapshots | 56 each | Maps and interactive filters |

# API collection methodology

1. Load the official static v4 response and normalize JSON.
2. Keep single-core, single-payload launches through 13 November 2020.
3. Select Falcon 9 and reindex 90 flights.
4. Derive landing outcomes and landing-pad IDs from core records.
5. Join the official enrichment snapshot and validate dates and outcomes.

Live enrichment endpoints returned 522/timeouts. The notebook explicitly documents the snapshot fallback.

https://github.com/rquilo/spacex-capstone-analysis/blob/main/notebooks/01_api_collection.ipynb

# Web scraping methodology

1. Download Wikipedia revision 1027686922.
2. Parse HTML with BeautifulSoup and find wikitable rows.
3. Retain numbered Falcon 9 rows, removing footnotes and normalizing whitespace.
4. Extract date, booster, site, payload, mass, orbit, customer and outcomes.
5. Export 121 records to CSV.

The later Wikipedia cutoff differs from the modeling snapshot.

https://github.com/rquilo/spacex-capstone-analysis/blob/main/notebooks/02_web_scraping.ipynb

# Data wrangling methodology

Inspect types, missing values and counts by launch site, orbit and landing outcome.

The supplied prepared snapshot already imputes missing payload mass with the mean. Keep the 26 missing landing-pad values as missing.

Derive Class from landing outcomes: 60 successes and 30 non-successes. Successful drone-ship landings account for 41 records.

GEO has 1 launch; GTO has 27. These orbit labels describe different categories.

https://github.com/rquilo/spacex-capstone-analysis/blob/main/notebooks/03_data_wrangling.ipynb

# Visual EDA methodology

Scatterplots compare flight number and payload with launch site and orbit. Colors identify the landing class.

Aggregate the mean of Class to calculate success rates by orbit and year. Display group sizes to avoid over-interpreting rare categories.

Encode categorical predictors for machine learning. Preserve the target outside the feature matrix.

https://github.com/rquilo/spacex-capstone-analysis/blob/main/notebooks/05_eda_visualization.ipynb

# SQL exploration methodology

Load the independent 101-row CSV into SQLite as SPACEXTBL.

Use DISTINCT, LIKE, SUM, AVG, MIN, GROUP BY, subqueries and date filters to answer ten questions.

Trim mission-outcome whitespace before counting. Keep uncertain payload status separate from unqualified success.

Distinguish mission success from first-stage landing success.

https://github.com/rquilo/spacex-capstone-analysis/blob/main/notebooks/04_eda_sql.ipynb

# Interactive analysis methodology

Folium: plot site coordinates, cluster launch records, color outcomes and calculate haversine distances to selected map features.

Dash: connect the site dropdown and payload RangeSlider to a callback that updates a success pie and payload scatterplot.

Verify three filter states in Chrome: all sites, KSC, and KSC with payload at most 6,000 kg.

https://github.com/rquilo/spacex-capstone-analysis/blob/main/notebooks/06_folium_maps.ipynb
https://github.com/rquilo/spacex-capstone-analysis/blob/main/dashboard.py

# Predictive analysis methodology

Split 90 rows into 72 training and 18 test observations (random_state=2).

Compare logistic regression, SVM, decision tree and KNN with 10-fold GridSearchCV. Select hyperparameters on training data.

Fit scaling inside the training pipeline for the report. Retain the historical course baseline separately for reproducibility.

Evaluate accuracy, balanced accuracy and the confusion matrix.

https://github.com/rquilo/spacex-capstone-analysis/blob/main/notebooks/07_predictive_analysis.ipynb

# Feature matrix and target definition

The four-column encoding requested in the EDA exercise produces 80 features.

Expanding each Boolean into two indicators adds three columns, matching IBM's supplied 83-column modeling matrix.

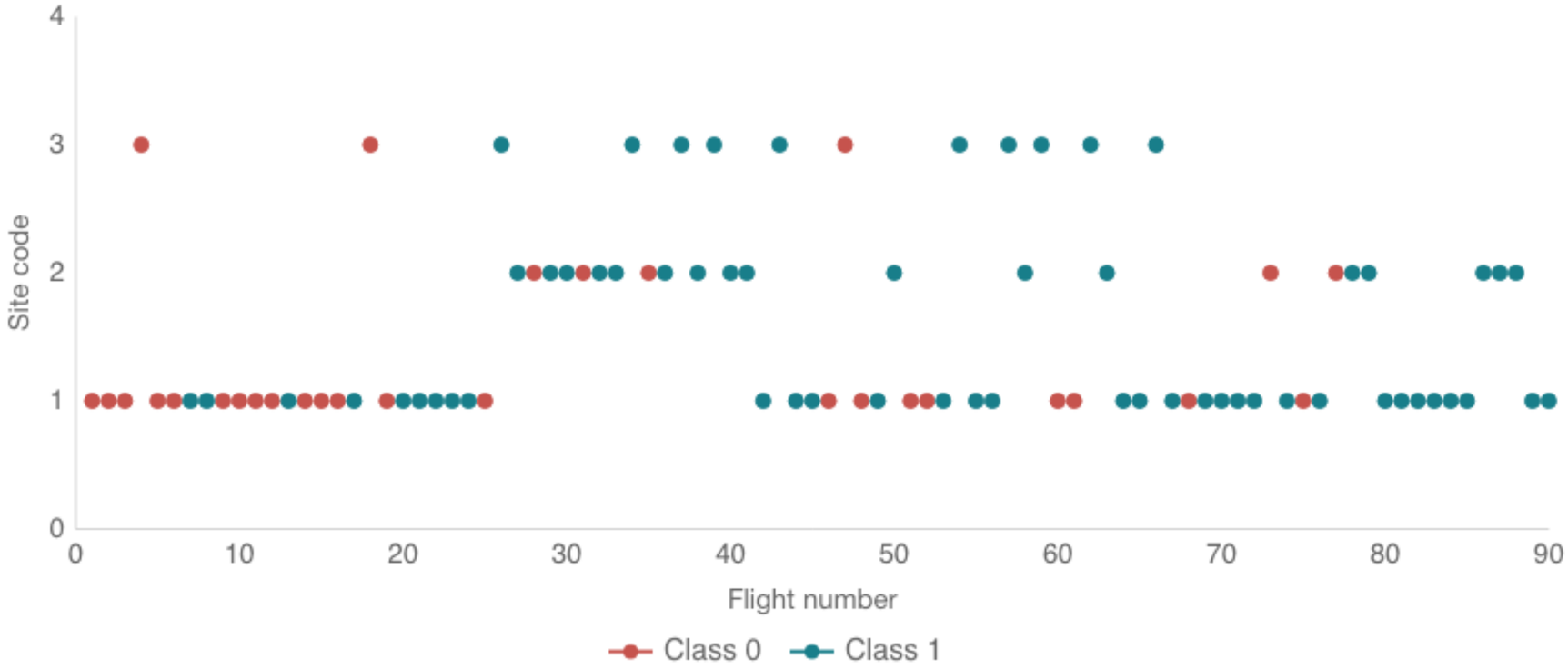
Outcome and Class are excluded from predictors. Flight number and booster serial can encode launch era, so random-split accuracy requires careful interpretation.

| Representation | Rows | Columns |
|---|---|---|
| Orbit, LaunchSite, LandingPad, Serial encoded | 90 | 80 |
| Also expand GridFins, Reused and Legs | 90 | 83 |

https://github.com/rquilo/spacex-capstone-analysis/blob/main/notebooks/05_eda_visualization.ipynb

# Flight number and launch site

Later flights contain more successes across all three modeling sites. This association also reflects changes over time.

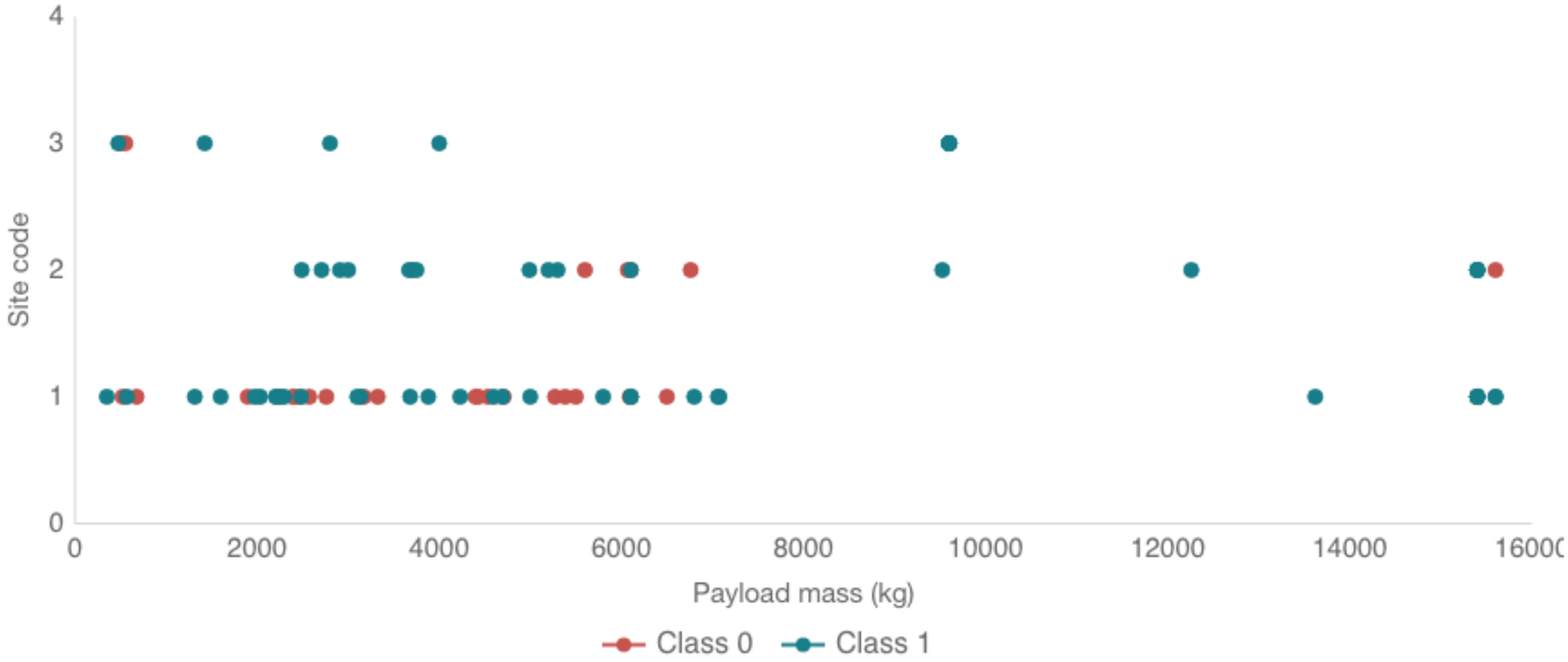



1 CCAFS SLC 40
2 KSC LC 39A
3 VAFB SLC 4E

https://github.com/rquilo/spacex-capstone-analysis/blob/main/notebooks/05_eda_visualization.ipynb

# Payload mass and launch site

Payload ranges differ across sites. Site effects are descriptive because mission types and launch era also vary.

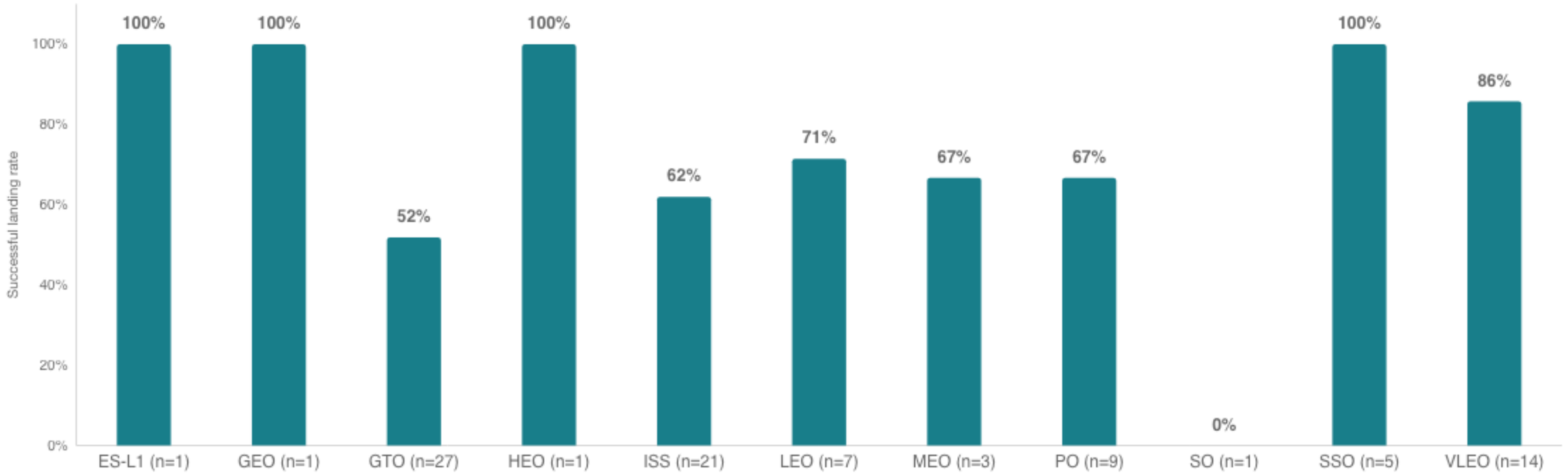



1 CCAFS SLC 40
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https://github.com/rquilo/spacex-capstone-analysis/blob/main/notebooks/05_eda_visualization.ipynb

# Landing success by orbit

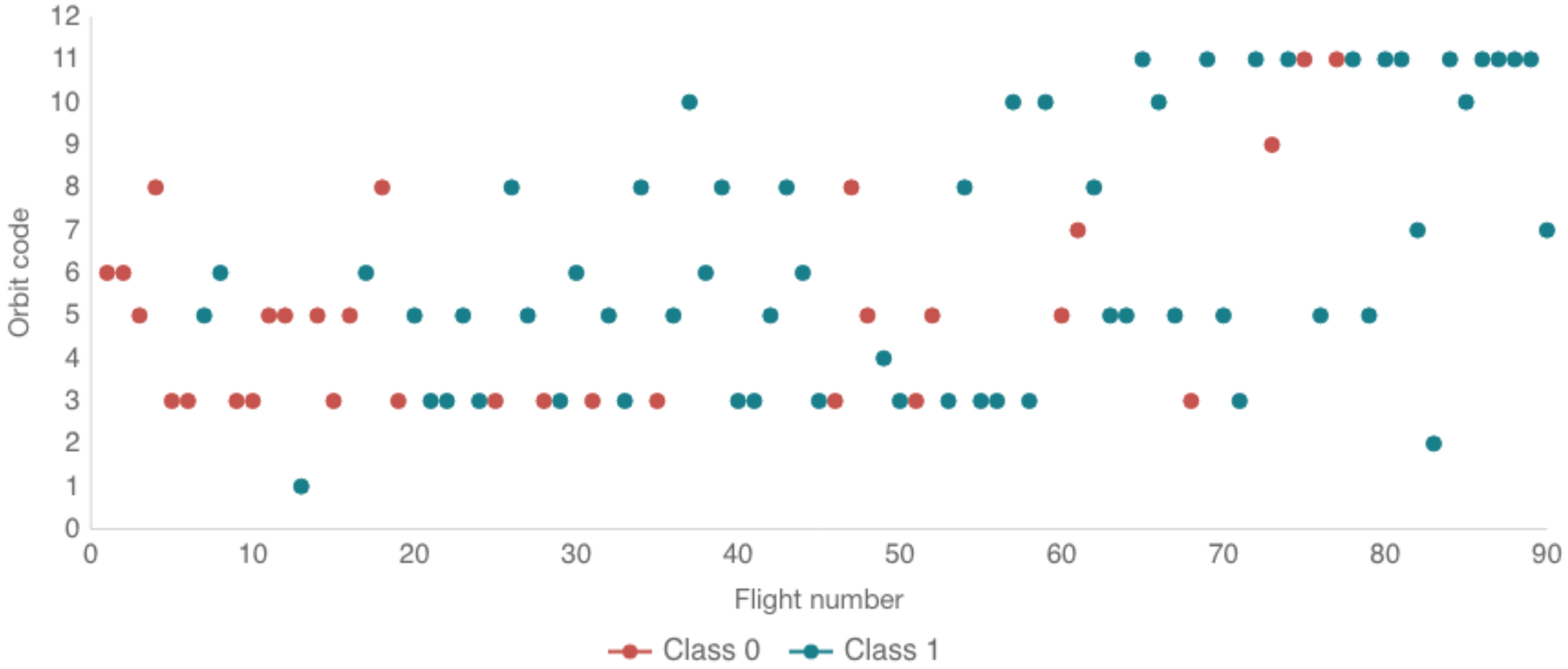
VLEO: 12/14 successful. GTO: 14/27. Several 100% groups contain only one observation.




https://github.com/rquilo/spacex-capstone-analysis/blob/main/notebooks/05_eda_visualization.ipynb

# Flight number and orbit

The mix of orbits changes over the launch sequence. Comparing early and late success rates without mission context can mislead.

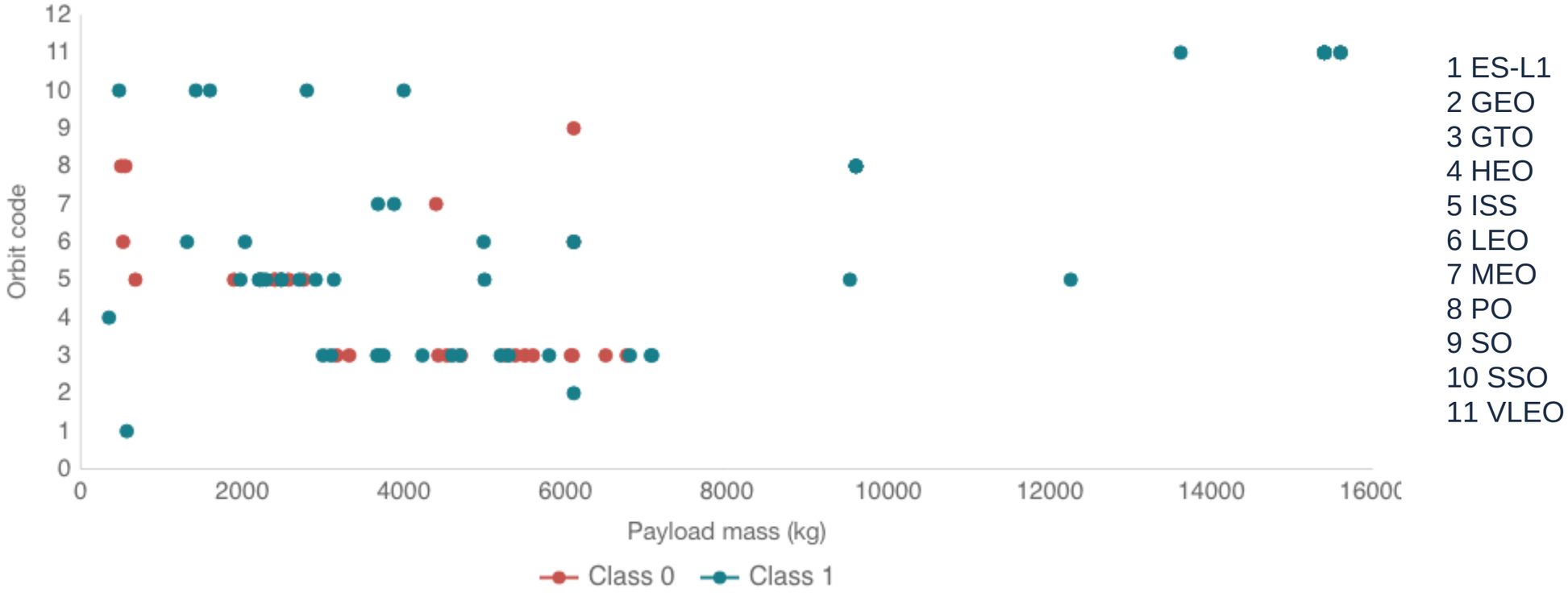



1 ES-L1
2 GEO
3 GTO
4 HEO
5 ISS
6 LEO
7 MEO
8 PO
9 SO
10 SSO
11 VLEO

https://github.com/rquilo/spacex-capstone-analysis/blob/main/notebooks/05_eda_visualization.ipynb

# Payload mass and orbit

Orbit and payload are related mission characteristics. The same payload mass does not imply the same landing conditions.

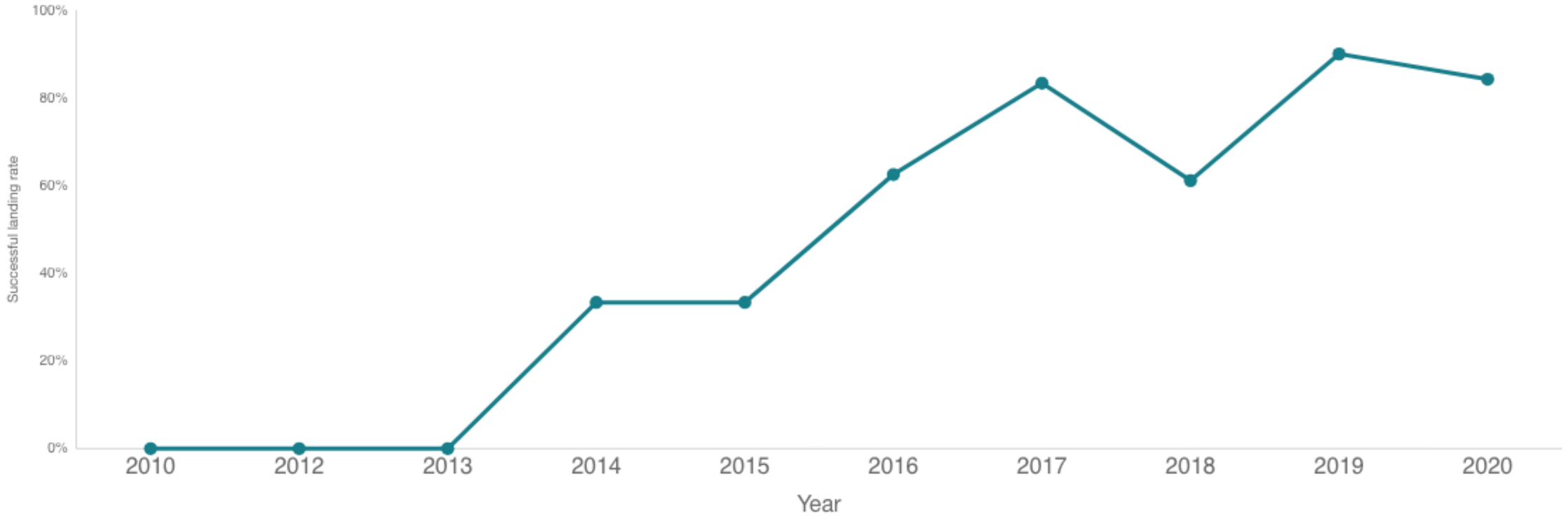



1 ES-L1
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# Success rate across years

2017: 15/18 successful. 2018: 11/18. 2019: 9/10. 2020: 16/19. Improvement is not monotonic.




https://github.com/rquilo/spacex-capstone-analysis/blob/main/notebooks/05_eda_visualization.ipynb

# SQL 1: Unique launch sites

```
SELECT DISTINCT Launch_Site FROM SPACEXTBL ORDER BY Launch_Site
```

| Launch site |
| --- |
| CCAFS LC-40 |
| CCAFS SLC-40 |
| KSC LC-39A |
| VAFB SLC-4E |

https://github.com/rquilo/spacex-capstone-analysis/blob/main/notebooks/04_eda_sql.ipynb

# SQL 2: First five CCA launches

SELECT Date, Booster_Version, Launch_Site FROM SPACEXTBL WHERE Launch_Site LIKE 'CCA%' ORDER BY Date LIMIT 5

| Date | Booster version | Launch site |
|---|---|---|
| 2010-06-04 | F9 v1.0 B0003 | CCAFS LC-40 |
| 2010-12-08 | F9 v1.0 B0004 | CCAFS LC-40 |
| 2012-05-22 | F9 v1.0 B0005 | CCAFS LC-40 |
| 2012-10-08 | F9 v1.0 B0006 | CCAFS LC-40 |
| 2013-03-01 | F9 v1.0 B0007 | CCAFS LC-40 |

https://github.com/rquilo/spacex-capstone-analysis/blob/main/notebooks/04_eda_sql.ipynb

# SQL 3: NASA CRS total payload

```
SELECT SUM(PAYLOAD_MASS__KG_) AS Total_kg FROM SPACEXTBL WHERE Customer = 'NASA (CRS)'
```

| Total payload (kg) |
| --- |
| 45596 |

https://github.com/rquilo/spacex-capstone-analysis/blob/main/notebooks/04_eda_sql.ipynb

# SQL 4: F9 v1.1 average payload

```
SELECT AVG(PAYLOAD_MASS__KG_) AS Average_kg FROM SPACEXTBL WHERE Booster_Version LIKE 'F9 v1.1%'
```

| Average payload (kg) |
| --- |
| 2,534.67 |

https://github.com/rquilo/spacex-capstone-analysis/blob/main/notebooks/04_eda_sql.ipynb

# SQL 5: First successful ground landing

```
SELECT MIN(Date) AS First_date FROM SPACEXTBL WHERE Landing_Outcome = 'Success (ground pad)'
```

| First date |
| --- |
| 2015-12-22 |

https://github.com/rquilo/spacex-capstone-analysis/blob/main/notebooks/04_eda_sql.ipynb

# SQL 6: Drone success, payload 4000 to 6000 kg

```
SELECT Date,Booster_Version,PAYLOAD_MASS__KG_ FROM SPACEXTBL WHERE
Landing_Outcome='Success (drone ship)' AND PAYLOAD_MASS__KG_>4000 AND
PAYLOAD_MASS__KG_<6000 ORDER BY Date
```

| Date | Booster version | Payload (kg) |
|---|---|---|
| 2016-05-06 | F9 FT B1022 | 4696 |
| 2016-08-14 | F9 FT B1026 | 4600 |
| 2017-03-30 | F9 FT B1021.2 | 5300 |
| 2017-10-11 | F9 FT B1031.2 | 5200 |

https://github.com/rquilo/spacex-capstone-analysis/blob/main/notebooks/04_eda_sql.ipynb

# SQL 7: Mission outcomes

```
SELECT TRIM(Mission_Outcome), COUNT(*) FROM SPACEXTBL GROUP BY TRIM(Mission_Outcome)
```

| Mission outcome | Missions |
|---|---|
| Success | 99 |
| Success (payload status unclear) | 1 |
| Failure (in flight) | 1 |

https://github.com/rquilo/spacex-capstone-analysis/blob/main/notebooks/04_eda_sql.ipynb

# SQL 8: Boosters carrying maximum payload (1/2)

SELECT Date,Booster_Version,PAYLOAD_MASS__KG_ FROM SPACEXTBL WHERE PAYLOAD_MASS__KG_=(SELECT MAX(PAYLOAD_MASS__KG_) FROM SPACEXTBL) ORDER BY Date

| Date | Booster version | Payload (kg) |
|---|---|---|
| 2019-11-11 | F9 B5 B1048.4 | 15600 |
| 2020-01-07 | F9 B5 B1049.4 | 15600 |
| 2020-01-29 | F9 B5 B1051.3 | 15600 |
| 2020-02-17 | F9 B5 B1056.4 | 15600 |
| 2020-03-18 | F9 B5 B1048.5 | 15600 |
| 2020-04-22 | F9 B5 B1051.4 | 15600 |

https://github.com/rquilo/spacex-capstone-analysis/blob/main/notebooks/04_eda_sql.ipynb

# SQL 8: Boosters carrying maximum payload (2/2)

```
SELECT Date,Booster_Version,PAYLOAD_MASS__KG_ FROM SPACEXTBL WHERE
PAYLOAD_MASS__KG_=(SELECT MAX(PAYLOAD_MASS__KG_) FROM SPACEXTBL) ORDER BY Date
```

| Date | Booster version | Payload (kg) |
|---|---|---|
| 2020-06-04 | F9 B5 B1049.5 | 15600 |
| 2020-09-03 | F9 B5 B1060.2 | 15600 |
| 2020-10-06 | F9 B5 B1058.3 | 15600 |
| 2020-10-18 | F9 B5 B1051.6 | 15600 |
| 2020-10-24 | F9 B5 B1060.3 | 15600 |
| 2020-11-25 | F9 B5 B1049.7 | 15600 |

https://github.com/rquilo/spacex-capstone-analysis/blob/main/notebooks/04_eda_sql.ipynb

# SQL 9: Failed drone landings in 2015

```
SELECT substr(Date,6,2) AS Month,Booster_Version,Launch_Site,Landing_Outcome FROM SPACEXTBL WHERE Landing_Outcome='Failure (drone ship)' AND substr(Date,1,4)='2015' ORDER BY Date
```

| Month | Booster version | Launch site | Landing outcome |
|---|---|---|---|
| January | F9 v1.1 B1012 | CCAFS LC-40 | Failure (drone ship) |
| April | F9 v1.1 B1015 | CCAFS LC-40 | Failure (drone ship) |

https://github.com/rquilo/spacex-capstone-analysis/blob/main/notebooks/04_eda_sql.ipynb

# SQL 10: Landing outcomes ranked 2010 to 2017 (1/2)

```
SELECT Landing_Outcome,COUNT(*) AS Outcomes FROM SPACEXTBL WHERE Date BETWEEN '2010-06-04' AND '2017-03-20' GROUP BY Landing_Outcome ORDER BY Outcomes DESC,Landing_Outcome
```

| Landing outcome | Outcomes |
|---|---|
| No attempt | 10 |
| Failure (drone ship) | 5 |
| Success (drone ship) | 5 |
| Controlled (ocean) | 3 |
| Success (ground pad) | 3 |
| Failure (parachute) | 2 |

https://github.com/rquilo/spacex-capstone-analysis/blob/main/notebooks/04_eda_sql.ipynb

# SQL 10: Landing outcomes ranked 2010 to 2017 (2/2)

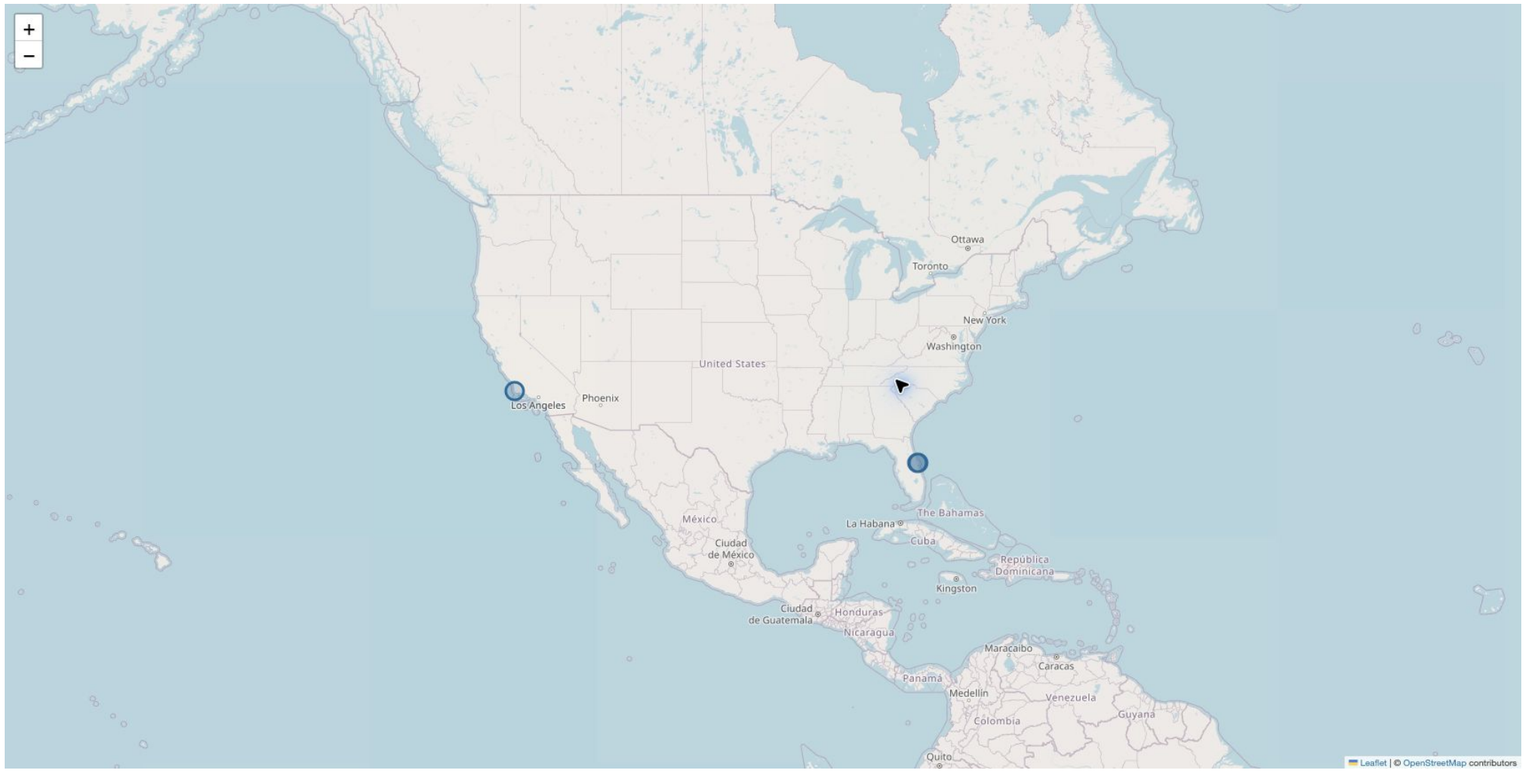
```
SELECT Landing_Outcome,COUNT(*) AS Outcomes FROM SPACEXTBL WHERE Date BETWEEN '2010-06-04' AND '2017-03-20' GROUP BY Landing_Outcome ORDER BY Outcomes DESC,Landing_Outcome
```

| Landing outcome | Outcomes |
|---|---|
| Uncontrolled (ocean) | 2 |
| Precluded (drone ship) | 1 |

https://github.com/rquilo/spacex-capstone-analysis/blob/main/notebooks/04_eda_sql.ipynb

# Launch-site locations

The 56-record geographic dataset spans Florida and California. LC-40 and SLC-40 labels refer to nearby positions in this historical source.




https://github.com/rquilo/spacex-capstone-analysis/blob/main/notebooks/06_folium_maps.ipynb

# Landing outcomes on the map

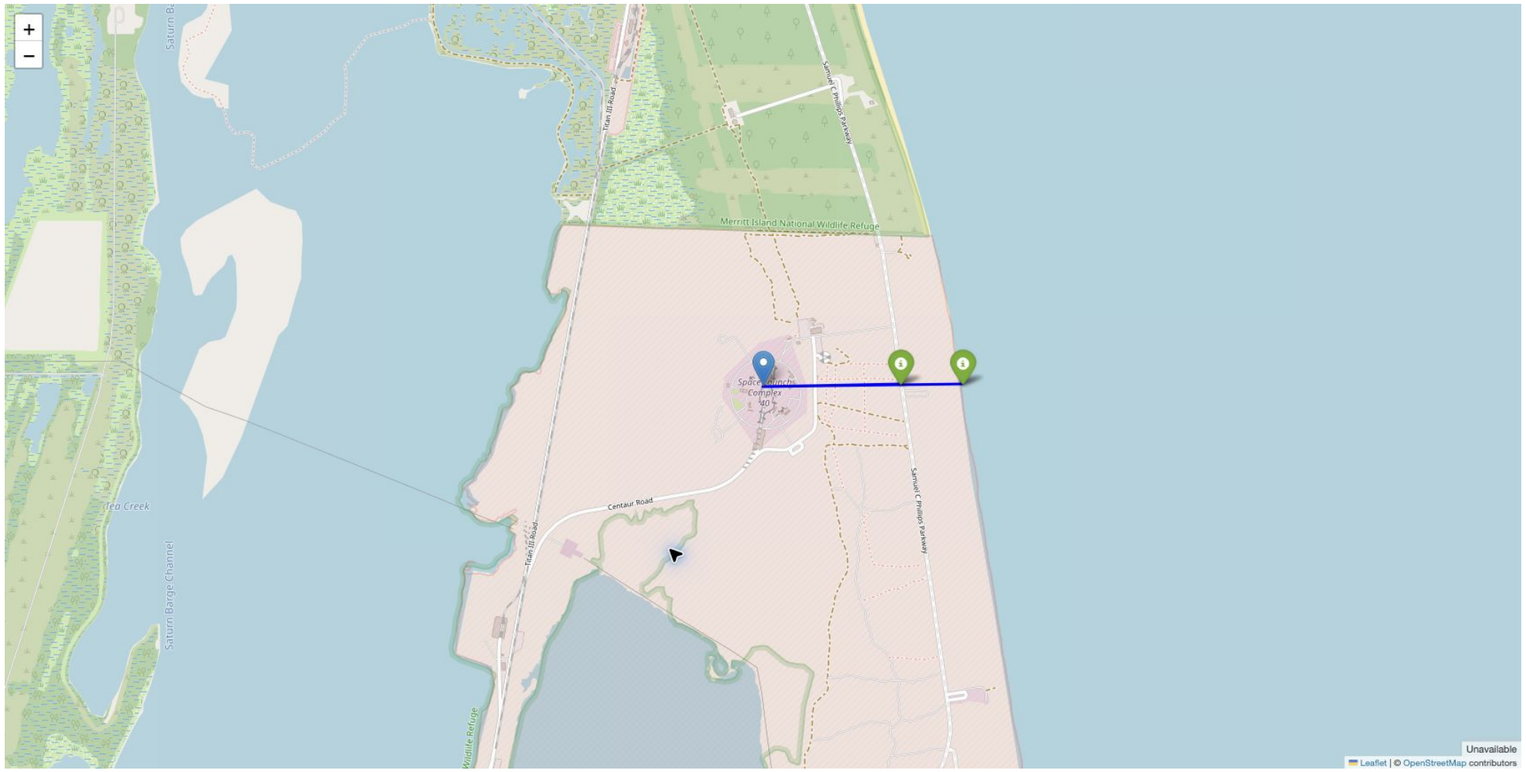
MarkerCluster expands coincident launch records. Green markers indicate Class 1 and red markers indicate Class 0. The view shows the LC-40 cluster.




https://github.com/rquilo/spacex-capstone-analysis/blob/main/notebooks/06_folium_maps.ipynb

# Proximity to coast and road

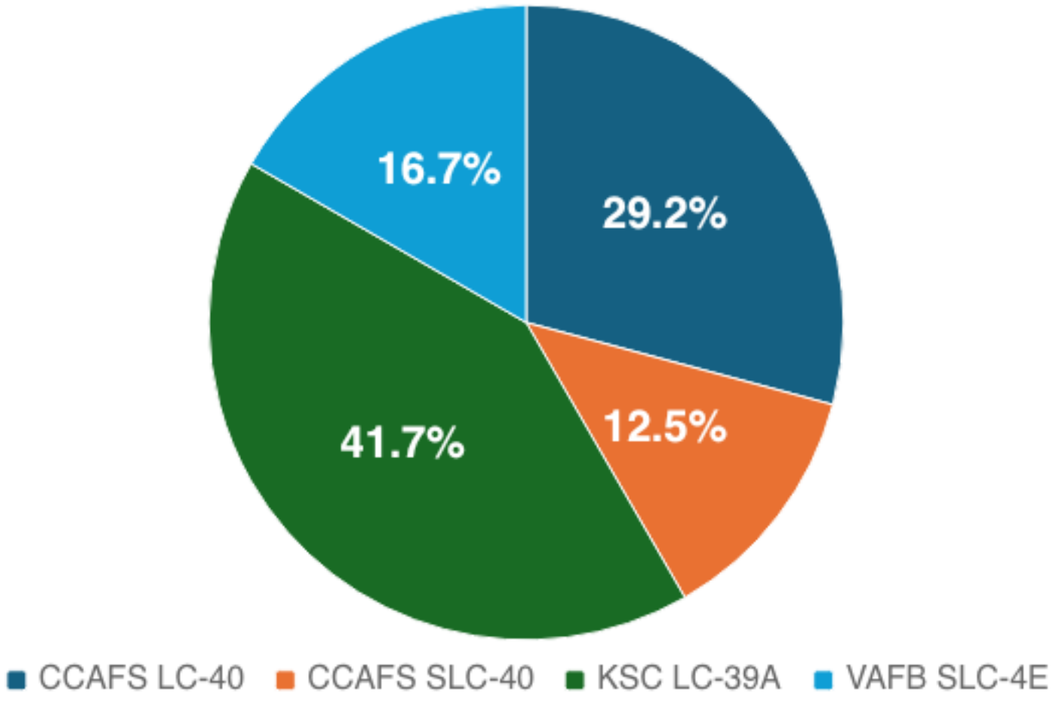
From CCAFS LC-40: 0.943 km to the selected coastline point and 0.650 km to Samuel C. Phillips Parkway. Haversine distances are approximate straight-line measurements.




https://github.com/rquilo/spacex-capstone-analysis/blob/main/notebooks/06_folium_maps.ipynb

# Dashboard: share of successful landings

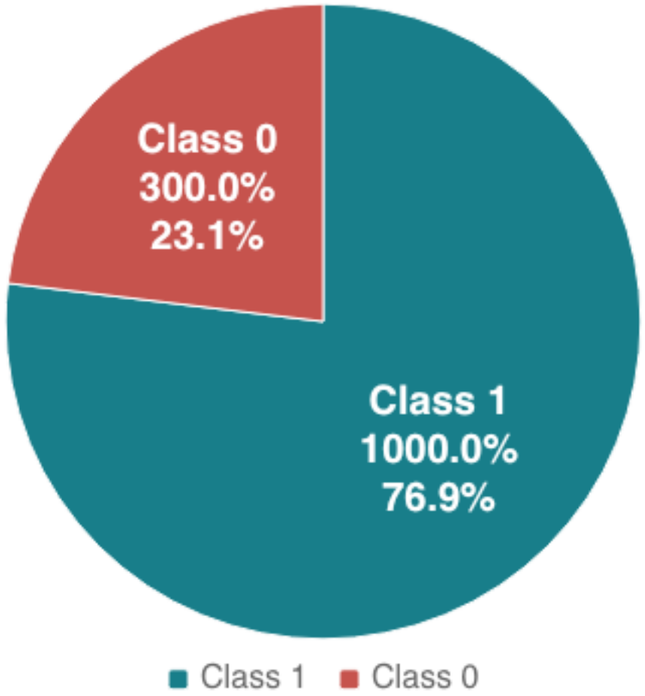
KSC LC-39A contributes 10 of the 24 successful landings (41.7%). This pie shows share of successes, not each site's success probability.




https://github.com/rquilo/spacex-capstone-analysis/blob/main/dashboard.py

# Dashboard: KSC has the highest success ratio

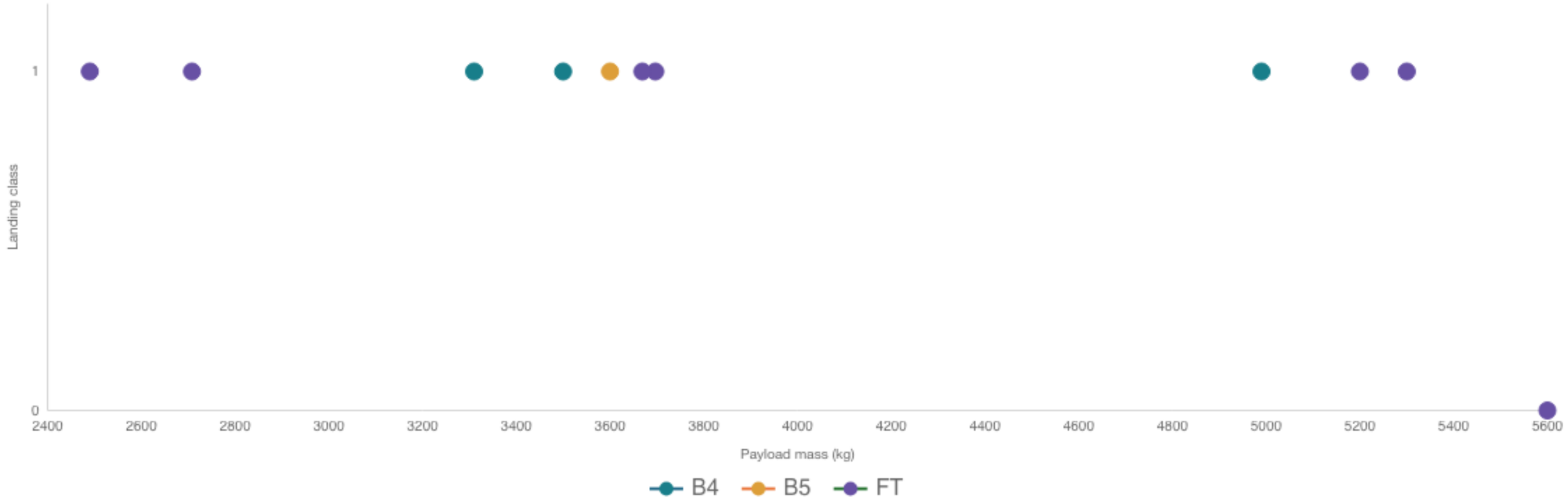
KSC LC-39A records 10 successes in 13 launches (76.9%) in the 56-row dashboard snapshot. The 90-row modeling snapshot has a different denominator.




https://github.com/rquilo/spacex-capstone-analysis/blob/main/dashboard.py

# Dashboard: payload filter changes the sample

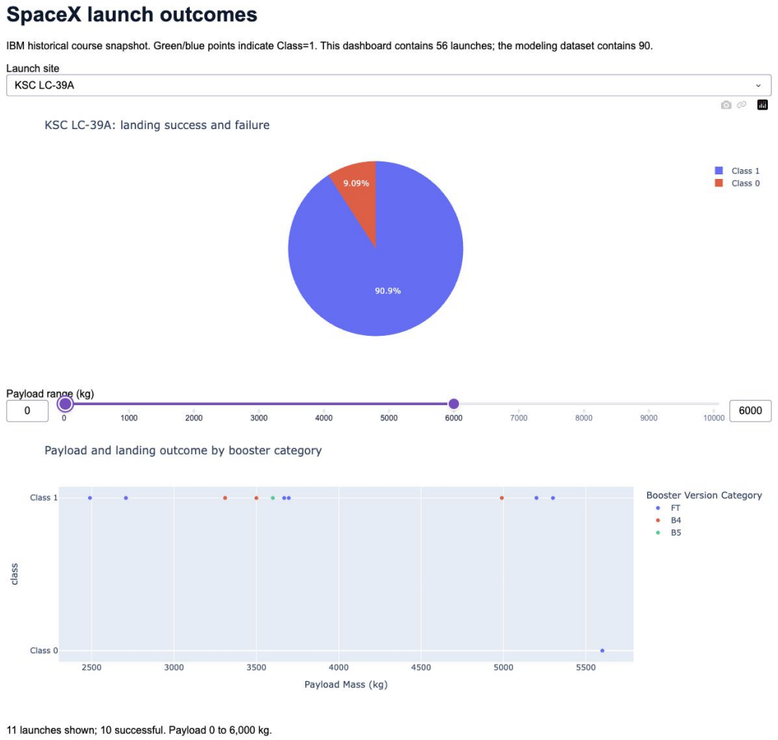
KSC with payload from 0 to 6,000 kg retains 11 launches and 10 successes. The observed ratio becomes 90.9%; filtering changes the population.




https://github.com/rquilo/spacex-capstone-analysis/blob/main/dashboard.py

# Working dashboard in Chrome

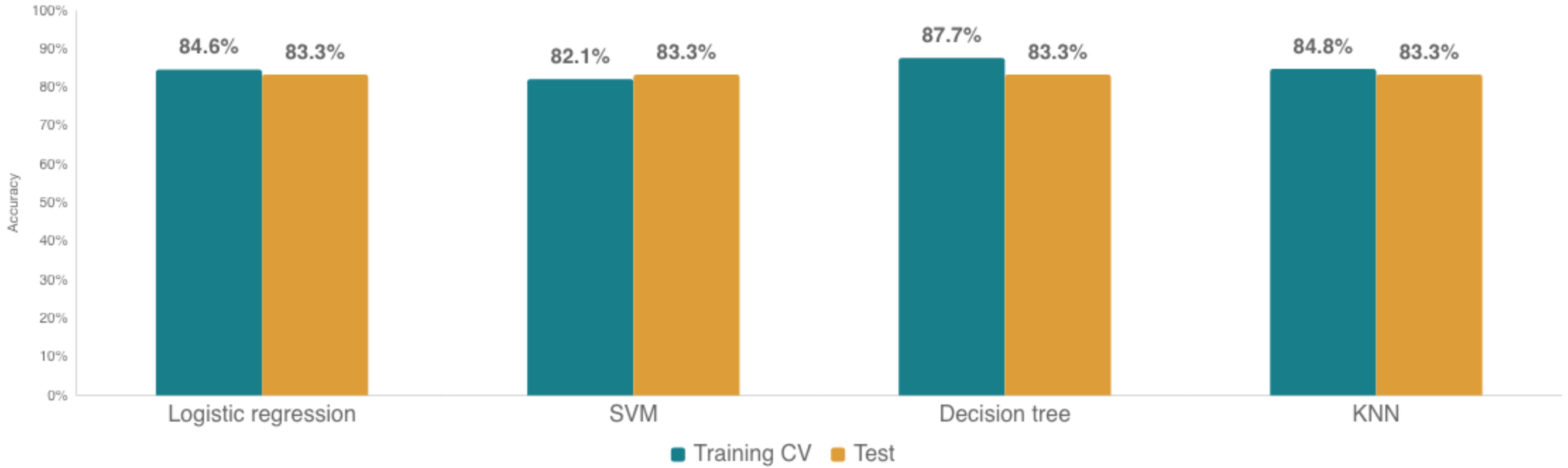
Verified the dropdown and RangeSlider callbacks against the underlying rows. Both charts update and the page reports the selected sample size.




https://github.com/rquilo/spacex-capstone-analysis/blob/main/dashboard.py

# Model comparison on a held-out test set

All models classify 15 of 18 test launches correctly. The tree has the highest mean training CV accuracy and is selected using that criterion.




https://github.com/rquilo/spacex-capstone-analysis/blob/main/notebooks/07_predictive_analysis.ipynb

# Selected hyperparameters and evaluation

Training-only preprocessing prevents the scaler from seeing validation or test observations. Each model has 75% balanced accuracy on this holdout.

| Model | Training CV | Test | Selected setting |
|---|---|---|---|
| Logistic regression | 84.64% | 83.33% | C=0.01, lbfgs |
| SVM | 82.14% | 83.33% | linear, C=0.0316 |
| Decision tree | 87.68% | 83.33% | gini, depth=8, leaf=2 |
| KNN | 84.82% | 83.33% | k=10, Manhattan |

https://github.com/rquilo/spacex-capstone-analysis/blob/main/notebooks/07_predictive_analysis.ipynb

# Decision tree confusion matrix

12 of 12 successful landings are identified. Three of six Class 0 cases are incorrectly predicted as successful.

Accuracy: 83.33%. Balanced accuracy: 75%. The majority-class baseline is 66.67%.

False positives matter when predicting safe reuse or planning mission economics.

| Actual / Predicted | Class 0 | Class 1 |
|---|---|---|
| Class 0 | 3 | 3 |
| Class 1 | 0 | 12 |

https://github.com/rquilo/spacex-capstone-analysis/blob/main/notebooks/07_predictive_analysis.ipynb

# Conclusions

Landing outcomes vary by launch era, site and orbit. In the modeling snapshot, KSC and VAFB have about 77% success versus 60% at CCAFS.

The selected tree reaches 83.3% holdout accuracy, but the small sample and three false positives limit operational use.

Maps and the dashboard support exploration of geographic context and filtered populations. They do not establish causal effects.

https://github.com/rquilo/spacex-capstone-analysis

# Limitations and next steps

A one-record change in the test set moves accuracy by 5.56 percentage points.

Historical serial and reuse metadata can contain hindsight. Rebuild timestamp-valid features and validate chronologically before deployment.

Separate failed landings from unattempted or unknown outcomes. Report recovery and reuse as distinct targets.

Expand the dataset and add uncertainty estimates. No cost or operational safety claim follows from this exercise.

https://github.com/rquilo/spacex-capstone-analysis/blob/main/notebooks/07_predictive_analysis.ipynb

# Reproducibility and repository

Eight executed notebooks, a runnable Dash application, interactive Folium maps, source snapshots and machine-readable results accompany this report.

Python 3.12. Install requirements.txt, execute notebooks in order, then run python dashboard.py.

AI-assisted preparation is disclosed in the notebooks and README. Sources retain their original rights.

https://github.com/rquilo/spacex-capstone-analysis